

Reading Concordances with Algorithms: a training course in key corpus linguistics methodology

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being in some square, the other exhibitions were IN houses, old shops, small factory spaces, offices and
might be described as 'wet' or 'dry' (W or D) and IN a humid temperate climate such as Britain expect
a powerful message put across by the curriculum IN most schools is that knowledge can be divided into
as like the trouble we had half an hour ago, when IN fact there were three lines, sections of which tran
opportunities for our **READING** arise naturally IN the course of the day; for instance, in the activity
traditional hand-copied **CONCORDANCES** IN several ways. One is that they will be much more
about Milan. 'How old were you when you arrived IN Italy?' The American journalist broke into his thou
of Modern Art, keen in discerning what was good IN **THE** arts of many ages and styles. As for writers
who felt it necessary to take aside both captains IN **no** 21ST minute and warn them about each team
region has undergone several media revolutions IN the last **CENTURY**. In principle, now that over
to some extent the ownership of the biggest hits IN each year), the aggregate market share of the top
focus from arts education in general, to one area IN particular, namely English lessons and the fiction
the horse was, shall I say dreamed up, erm and IN fact we had one or two horses, Clydesdale horse
about woods. It was merely that he never wanted IN a tournament round to risk anything which might

What we have learned so far

- Concordances collect occurrences of a node for “vertical reading”
- Help (corpus) linguists to identify recurrent patterns
- It is not always easy to spot patterns in concordances (except for educational and famous examples)

Recap: Theoretical foundations

- Lexicogrammatical level:
 - Lexical item
 - Patterns (as in Pattern Grammar)
- Textual level
 - Local textual functions
 - Lexical Priming (Hoey 2005)
- Purpose of the analysis:
 - The study of fiction requires a “fiction-specific” approach, different from analysing socio-political debate around environmental issues

Methodology: a reading procedure (Sinclair 2003)



Step	Summary
1. Initiate	Decide on the strongest patterns to the left or right of the node (e.g., recurring words, word forms, or parts of speech.)
2. Interpret	Find a hypothesis that links (most of) the repeated words. The link could be lexical, semantic, or syntactic.
3. Consolidate	Look for further evidence to support the hypothesis (e.g., similar meanings expressed in different ways, words in more distant positions.)
4. Report	Write out an explicit, testable version for the future.
5. Recycle	Repeat steps 1–4 with the next most important pattern. Continue until you cannot find any more repeated patterns. Classify the remaining lines as either unusual or a potential pattern that is rare in this set of lines.
6. Result	Make a list of final hypotheses.
7. Repeat	Repeat with a new sample to confirm, extend, or revise your hypotheses.

Concordancing tools

- Are created for different purposes
- Follow different traditions (cf. yesterday's origin stories)
- Use different terminology (cf. concordance vs contexts)
- Combine functionalities (key words, n-grams, etc.)
- Can be specialised or general-purpose
- Can be web-based or standalone
- Often come with corpora
- Are continuously developed (until they aren't)
- Provide user support (hopefully)

Concordancing tools: CQPweb

Your query "[word="what"%c & !bound(s)] "a"%c [pos="JJ.*"]+ "night"%c" returned 18 matches in 15 different texts (in 98,511,777 words [9,802 texts]; frequency: 0.18 instances per million words)

[4.616 seconds]

Navigation controls: | < << >> > | Show Page: 1 Line View Show in random order New query Go!

No	Filename	Solution 1 to 18 Page 1 / 1		
1	film_1944_the_climax	My dearJarmila ,	what a great night	for you .
2	film_1956_the_rainmaker	oes crazy on a hot night , and maybe that 's	what a hot night	is for .
3	film_1962_experiment_in_terror	Wow ,	what a miserable night	.
4	film_1966_frankie_and_johnny	tee that you 'll have a ball Come one and all	What a great night	you 've got in store You 'll wanna keep comi
5	film_1966_frankie_and_johnny	e Hey , everyone , let 's go on with the show	What a great night	you 've got in store I 'll bet you 'll wanna kee
6	film_1970_julius_caesar	I am glad on ` t.	What a fearful night	is this !
7	film_1970_the_aristocats	Oh , dear ,	what a terrible night	.
8	film_1973_robin_hood	it 's gone Love goes on and on Oh , Robin ,	what a beautiful night	.
9	film_1985_weird_science	Crimson morning skyline Whoa oh	What a weird night	, huh ?
10	film_1989_dead_bang	It 's a wonder	what a good night	's sleep will do for you .
11	film_1990_gremlins_2_the_new_batc	When they think it's sunset and see	what a nice night	it is... they 'll muster in the lobby

CQPweb: A hands-on example

- Goal: Understand this [→] guy's rhetoric, terminology, phraseology and topics
- We need a large collection of texts written by Trump, e.g. the **Trump Twitter Archive** [www.thetrumparchive.com]
- Step 1: Compile a linguistic corpus ✓
- Step 2: Upload to indexing & search engine ✓
- Step 3: Analyse through concordance reading



CQPweb: A hands-on example

- CQPweb server @ FAU:
<https://corpora.linguistik.uni-erlangen.de/cqpweb/>
- Login: **student1** ... **student15** | password: ●●●●●●●●

A screenshot of the CQPweb login interface. The page has a light blue header with a 'CL' logo on the left, a welcome message in the center, and a 'CWB' logo on the right. The main content area contains a login form with fields for 'Enter your username:' and 'Enter your password:', a 'Stay logged in on this computer:' checkbox, and two buttons: 'Click here to log in' and 'Clear form'. At the bottom, there are links for 'Create account' and 'Full account-control options', and a footer section titled 'Corpora available on this server'.

Welcome to *CQPweb* at FAU Erlangen-Nürnberg!

This server is maintained by the
Chair of Computational Corpus Linguistics

Enter your username:

Enter your password:

Stay logged in on this computer: ☐

[Click here to log in](#) [Clear form](#)

[Create account](#) | [Full account-control options](#)

Corpora available on this server

Simple search

Menu	Trump Twitter Archive: <i>powered by CQPweb</i>
Corpus queries	Standard Query
Standard query	make america great again
Restricted query	
Word lookup	
Frequency lists	
Keywords	
Analyse corpus	
Saved query data	
Query history	Query mode: Simple query (ignore case) Simple query language syntax
Saved queries	Number of hits per page: 50
Categorised queries	Match strategy: Standard
Upload a query	Restriction: None (search whole corpus)
Create/edit subcorpora	Start Query Reset Query
Corpus info	System messages
View corpus metadata	New Arrangement of Corpora
No corpus documentation available	2018-06-06 The corpora have been newly arranged. If you cannot find a corpus that belongs to you or lost your access privileges, please contact us. All corpora listed under "Uncategorised" will be deleted in the long run. Please let us know if there are any issues with this.
About CQPweb	
CQPweb main menu	
Your user page	
Help system	
Video tutorials	
Who did it?	

Concordance (kwic = key word in context)

Your query "make america great again" returned 585 matches in 585 different texts (in 1,332,122 words [56,570 texts]; frequency: 439.15 instances per million words), ordered randomly

[0.04 seconds]

No.	Text	Solution 1 to 50	Page 1 / 12
1	t1037417543017877500	Join me tomorrow night at 7:00 pm MDT in Billings , Montana for a	MAKE AMERICA GREAT AGAIN RALLY ! Get your tickets here : https://t.co/fyHduA2Peo
2	t0761622637235712000	upport to get to the White House and defeat #CrookedHillary . Let's	Make America Great Again ! https://t.co/u2WQondrgm
3	t0763860288533856300	ounting on your help to defeat Hillary Clinton and her cronies . Let's	Make America Great Again ! https://t.co/ISK0l0j29K
4	t0585269105068007400	ldtrump @cleopattrausa Donald you are the next Ronald Reagan .	Make America great again . "" Thank you .
5	t0620994157004357600	t . We are energized & amp , ready to take our country back . Let's	Make America Great Again ! https://t.co/u25yl5T7E8
6	t0697825190282182700	He (or she) who hesitates is lost :	MAKE AMERICA GREAT AGAIN !
7	t080741732610697825190282182700	, together , we will	MAKE AMERICA GREAT AGAIN ! Watch : https://t.co/EyLOo26FqW
8	t070021001310700210013	urday is a BIG day .	MAKE AMERICA GREAT AGAIN !
9	t047331056310473310563	realDonaldTrump ""	Make America great again !
10	t122721695811227216958	realDonaldTrump :	MAKE AMERICA GREAT AGAIN and then , KEEP AMERICA GREAT !
11	t082201665310822016653	we will , together ,	MAKE AMERICA GREAT AGAIN !
12	t056241491910562414919	Let's together	Make America Great Again ! Vote Trump at https://t.co/YoNf60s0Im
13	t0612415976890593300	J.S. is becoming a dumping ground for the world . Pols don't get it .	Make America Great Again !
14	t0685215573943279600	uld only get a small fraction of this 25k crowd in . The movement to	Make America Great Again is unbelievable ! https://t.co/NHPdGm57YJ
15	t0771115576381427700	r she didn't go to Mexico . She doesn't have the drive or stamina to	MAKE AMERICA GREAT AGAIN !
16	t0476110011047611001		MAKE AMERICA GREAT AGAIN ! MAKE AMERICA SAFE AGAIN !
17	t1226711459107150300	Thank you Allentown , Pennsylvania ! Together , we are going to	MAKE AMERICA GREAT AGAIN ! https://t.co/gsFSghkmdM https://t.co/aA0I8pBEGb https://t.co/ngL
18	t0616897546024304600	'heguhMantap : @realDonaldTrump you are next President USA ""	Make America Great Again !
19	t0603631285777596400	in your best interests . Time for the US gov't to do the same . Let's	Make America Great Again !
20	t1307675370322145300		MAKE AMERICA GREAT AGAIN !
21	t1247555316863553500	great American Patriots who voted for me . AMERICA FIRST . and	MAKE AMERICA GREAT AGAIN will have a GIANT VOICE long into the future . They will not be d

frequency
information

random ordering

[- click for context](#)

click / hover
for metadata

Extended context display with formatting

Displaying extended context for query match in text **t0697825190282182700**

Select action...

Go!

Show tags

2016-02-11 00:34:25 [👁️] 1279, ❤️ 4331

I was referring to the fact that Jeb Bush wants to keep common core .

2016-02-11 00:37:51 [👁️] 1184, ❤️ 4258

"" @HARyder : Which is it @realDonaldTrump ? Are you planning on getting rid of Common Core or keeping it ? Get rid of it fast .

2016-02-11 01:12:19 [👁️] 2927, ❤️ 8568

THANK YOU- Clemson , South Carolina ! #MakeAmericaGreatAgain #SCPrimary <https://t.co/FgACmaFxxc>

2016-02-11 01:51:40 [👁️] 5097, ❤️ 13386

I have been consistent in my opposition to Common Core . Get rid of Common Core --- keep education local !

2016-02-11 13:13:09 [👁️] 1640, ❤️ 5432

There are no buyers for the worthless @NYDailyNews but little Mort Zuckerman is frantically looking . It is bleeding red ink - a total loser !

2016-02-11 14:11:56 [👁️] 3469, ❤️ 9767

Jeb Bush spent more than \$ 40,000,000 in New Hampshire to come in 4 or 5 , I spent \$ 3,000,000 to come in 1st . Big difference in capability !

2016-02-11 14:40:04 [👁️] 1436, ❤️ 4032

. @MarkHalperin showed a focus group on @Morning_Joe me using a very bad word . I never said the word , left an open blank . Please apologize !

2016-02-11 15:14:10 [👁️] 2589, ❤️ 6342

Remember , it was the Republican Party , with the help of Conservatives , that made so many promises to their base , BUT DIDN'T KEEP THEM ! Hi DT

2016-02-11 15:51:04 [👁️] 2596, ❤️ 7190

He (or she) who hesitates is lost : **MAKE AMERICA GREAT AGAIN !** ← **this is the match**

2016-02-11 18:45:59 [👁️] 2277, ❤️ 7161

Jeb failed as Jeb ! He gave up and enlisted Mommy and his brother (who got us into the quicksand of Iraq) . Spent \$120 million . Weak-no chance !

2016-02-11 19:39:33 [👁️] 3659, ❤️ 9115

We are getting reports from many voters that the Cruz people are back to doing very sleazy and dishonest "" pushpolls "" on me . We are watching !

2016-02-11 22:03:54 [👁️] 1604, ❤️ 5195

Heading to Baton Rouge , Louisiana for a speech . Expecting a very large crowd ! See you soon . #Trump2016 #MakeAmericaGreatAgain

2016-02-11 22:19:33 [👁️] 1646, ❤️ 4445

"" @truthinvest : @CNN @tedcruz @realDonaldTrump Ted Cruz is the definition of sleaze . Just ask @RealBenCarson ""

2016-02-11 23:44:06 [👁️] 2478, ❤️ 5355

Cruz caught cold in lie after denial of push polls like lies w/ @RealBenCarson . How can he preach Christian values ? <https://t.co/p3yGL02ABA>

2016-02-12 00:06:30 [👁️] 2179, ❤️ 7547

Just landed in Baton Rouge , Louisiana . Reports are out that lines are three quarters of a mile to get in . Wow ! #MakeAmericaGreatAgain

2016-02-12 02:28:28 [👁️] 3822, ❤️ 9677

THANK YOU- Baton Rouge , Louisiana ! WE will #MakeAmericaGreatAgain ! #Trump2016 <https://t.co/XV7Ele7A2I>

2016-02-12 03:10:52 [👁️] 2110, ❤️ 5156

Weak JEB getting thrown out by management during speech . Do you think he will be this tough on Putin & amp , others ? <https://t.co/Tqej1euLVL>

2016-02-12 03:20:33 [👁️] 3357, ❤️ 9104

Lying Cruz put out a statement , " Trump &

Sorted concordance: *great*

action menu

Your query "[lemma="great"%cd]", in subcorpus "Originals", returned 7,332 matches in 6,771 different texts (in 1,084,300 words [43,878 texts]; frequency: 6,761.97 instances per million words), sorted on *position +1* (7,332 hits) [0.126 seconds - retrieved from cache]

[<](#)[<<](#)[>>](#)[>](#)

Show Page: 1

Line View

Show in random order

Go!

Sort control:

Position: 1 Right

Tag restriction: None

☐ exclude

Starting with:

☐ exclude

- ✓ Choose action...
- New query
- Thin...
- Frequency breakdown
- Distribution
- Dispersion
- Sort
- Collocations...
- Download...
- Categorise...
- Save current query result...

No.	Text	Solution 2101 to 2150	Page 43 / 147
2101	t1122270956192272400	Sincerest THANK YOU to our great	Border Patrol Agent who sto
2102	t1104017664131907600	tending record numbers of illegal immigrants - but we need the Wall to help our great	Border Patrol Agents !
2103	t1075954128222871600	ans voted and won , 217-185 . Nancy does not have to apologize . All I want is GREAT	BORDER SECURITY !
2104	t0876899250071904300	 getting great	border security and healthcare . #VoteRalphNorman tomorrow !
2105	t1024248479386923000	One of the reasons we need Great	Border Security is that Mexico's murder rate in 2017 increased by 27% to 31,1
2106	t0977855968364171300	ild our Military , many jobs are created and our Military is again rich . Building a great	Border Wall , with drugs (poison) and enemy combatants pouring into our Co
2107	t0968704442110545900	Administration and rejected the attempt to stop the government from building a great	Border Wall on the Southern Border . Now this important project can go forwar
2108	t0830405706255913000	I am reading that the great	border WALL will cost more than the government originally thought , but I have
2109	t0741286391200505900	The great	boxing promoter , Don King , just endorsed me . Nice !
2110	t1198679802913415200	Agnes , your great	boy Ronald is looking down , very proud of you ! https://t.co/BHPu6IIdAN
2111	t0564215204154966000	ldTrump @Newsmax_Media @dpatten32 Time for Trump to take charge of the greatest	brand in the world , the USA ! ""
2112	t0263720538923470850	's acquisition of Lucas Film is a smart deal for both sides . Disney just bought a great	brand which will keep producing revenue .
2113	t0862283792559616000	years , as a pol in Connecticut , Blumenthal would talk of his great	bravery and conquests in Vietnam - except he was never there . When
2114	t0334834807483818000	I totally respect that Angelina Jolie has shown such great	bravery in the face of danger - she has really come a long and positive way !
2115	t1060515116700045300	 Great	bravery shown by police . California Highway Patrol was on scene within 3 mir
2116	t0979082457340407800	Great	briefing this afternoon on the start of our Southern Border WALL ! https://t.co/p
2117	t0746458701565988900	Many people are equating BREXIT , and what is going on in Great	Britain , with what is happening in the U.S. People want their country back !
2118	t0800887087780294700	Many people would like to see @Nigel_Farage represent Great	Britain as their Ambassador to the United States . He would do a great job !
2119	t0580623025827881200	You rocked both events . The @washingtonpost knows it too !! Thank you for a great	brunch and a great talk !!

Frequency breakdown: *fake*

full process history

Query "{fake}", in subcorpus "Originals", returned 1,194 matches in 1,117 different texts (in 1,084,300 words [43,878 texts]; frequency: 1,101.17 instances per million words)

Showing frequency breakdown of words in this query, at position 1 to the Right; there are 104 different types and 1,194 tokens at this concordance position.

[0.006 seconds - retrieved from cache]

<

<<

>>

>

Breakdown position: 1 Right

Frequency breakdown of words only

Go!

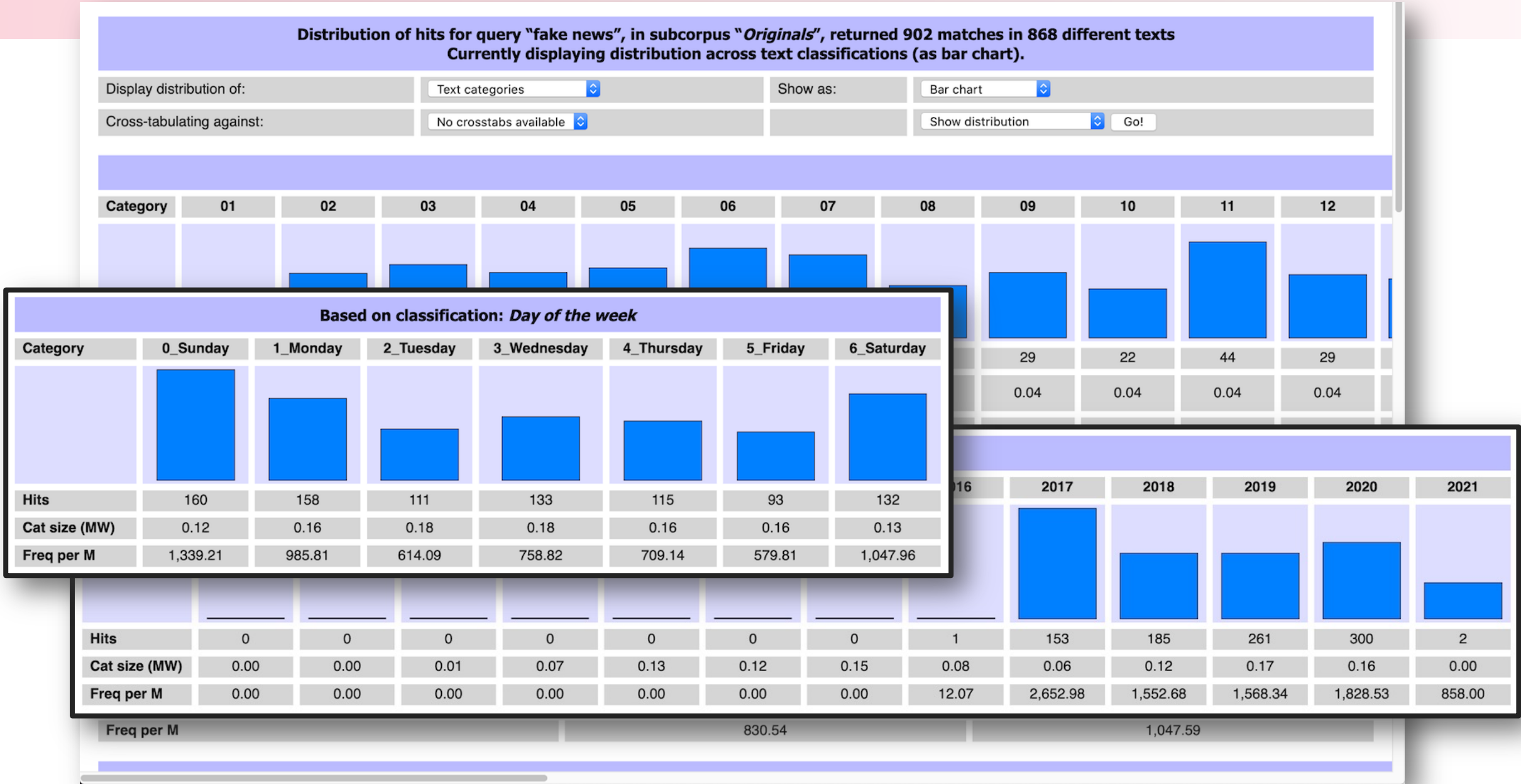
No.	Query result	No. of occurrences	Percent
1	NEWS	902	75.54%
2	and	24	2.01%
3	Dossier	21	1.76%
4	media	19	1.59%
5	&	17	1.42%
6	,	15	1.26%
7	!	9	0.75%
8	)	9	0.75%
9	Polls	7	0.59%
10	reporting	7	0.59%
11	Story	7	0.59%
12	Whistleblower	7	0.59%
13	Suppression	6	0.5%
14	.	5	0.42%
15	ballots	5	0.42%
16	(	4	0.34%
17	""	3	0.25%
18	@NBCNews	3	0.25%
19	ad	3	0.25%

Time for a closer look: *fake news*

Menu	
Corpus queries	
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Trump Twitter Archive: <i>powered by CQPweb</i>	
Standard Query	
fake news	
Query mode:	Simple query (ignore case) Simple query language syntax
Number of hits per page:	50
Match strategy:	Standard
Restriction:	Subcorpus: Originals (1,084,300 words in 43,878 texts)
Start Query Reset query	
System messages 	
New Arrangement of Corpora	
2018-06-06	The corpora have been newly arranged. If you cannot find a corpus that belongs to you or lost your access privileges, please contact us. All corpora listed under "Uncategorised" will be deleted in the long run. Please let us know if there are any issues with this.

Distribution across metadata categories



Collocation analysis

key parameters

Collocation controls

Collocation based on:

Lemma

Collocation window from:

5 to the Left

Freq(node, collocate) at least:

5

Filter results by:

specific collocate:

Statistic:

Log-likelihood

Collocation window to:

5 to the Right

Freq(collocate) at least:

5

and/or tag:

V

V

Submit changed parameters

Go!

Extra information:

Log-likelihood (LL) scores collocations by significance: the higher the score, the more evidence you have that the association is not due to chance. More frequent words tend to get higher log-likelihood scores, because there is more evidence for such words.

There are 1,497 different lemmas in the collocation database for this query (Query "fake news", in subcorpus "Originals", returned 902 matches in 868 different texts)

[0.175 seconds - retrieved from cache]

No.	Lemma	Total no. in this subcorpus	Expected collocate frequency	Observed collocate frequency	In no. of texts	Log-likelihood
1	report	615	5.116	39	39	92.712
2	cover	156	1.298	18	18	63.171
3	refuse	176	1.464	17	17	53.748
4	be	35,694	296.929	423	361	49.536
5	doesn't	249				
6	mention	129				
7	say	1,987				
8	talk	677				
9	refer	52				
10	hate	242				
11	try	548				
12	go	2,877	23.933	45	42	14.898

click for examples

Choose settings for proximity-based collocations:

Include annotation:

Lemma

Include

Exclude

POS-disambiguated lemma

Include

Exclude

Part of speech

Include

Exclude

Maximum window span:

+ / -

5

Create collocation database

CQPweb exercise

- Activate corpus **ParlSpeech UK**
 - corpus of debates of the UK parliament 1988–2019
 - Rauh, C. & Schwalbach, J. (2020). The ParlSpeech V2 data set: Full-text corpora of 6.3 million parliamentary speeches in the key legislative chambers of nine representative democracies. <https://doi.org/10.7910/DVN/L4OAKN>
 - ~ 2 M turns with a total of ~ 400 M tokens
- Concordance analysis for query: **water and sanitation**
 - what recurrent patterns can you identify?
 - can you find evidence for larger units of meaning?

Coffee break



What we have learned so far from using CLiC & CQPweb?

- ☐ What strategies have you applied?
- ☐ How do you describe what you do?

Practices in different disciplines: survey of strategies

- 10 articles / case studies per discipline (40 total)
 - Lexicography
 - Data-driven learning
 - Corpus-assisted discourse studies
 - Literary stylistics
 - post-2000 publication; concrete examples
 - not included: studies using parallel or learner corpora
- Content analysis of methods sections (not literature review)
- Challenge: no systematic standards for describing methodology, mix of tool and method

Selecting

Sampling

A random **sample** of 100 concordances of each pair was, subsequently, **closely examined** for semantic prosody. **(Lexicography)**

Filtering

A concordance **query** for *carry out* with *research* in the context of three tokens to the left and to the right returned 160 hits, and most lines indeed exhibited the collocation *carry out research*, as exemplified in Figure 8, **(Lexicography)**

Selecting

Cleaning

As we were interested in lexical compounds linked to carbon dioxide rather than carbon as an element, such lexical combinations as, for example, *carbon molecule* or *carbon fibre* were removed. Nonsensical combinations as well as duplicates arising from differences in spelling such as *carbon offset* and *carbon-offset* were also eliminated. **(CADS)**

Threshold

We analysed the first 500 concordances of *seguire* chosen as “good examples” by the Sketch Engine (cf. Kilgarriff et al., 2008) **(Lexicography)**

Ordering

Sorting

WordSmith Tools was mainly used for **sorting** and categorising concordances because it allows the analyst to assign values to lines and to move easily from concordance to text. [Figure 3 shows sorting by 1R] **(CADS)**

The students were then instructed to search for the preceding underlined word *friendship* using IntelliText, and to examine the concordance lines to identify which preposition should follow *friendship* in this context, by **sorting** right or left contexts in keyword-in-context (KWIC) lines. **(DDL)**

Ordering

Ranking

We analysed the **first 500** concordances of *seguire* **chosen as “good examples”** by the Sketch Engine (cf. Kilgarriff et al., 2008)
(Lexicography)

Reading Concordances R21 Project

What do users do to organise concordance lines for interpretation? What drives the choices they make, beyond their intuition and the features of concordancing tools they are familiar with?

Our goals:

1. **Develop a consistent methodology** of concordance reading strategies and suggesting a transparent terminology that describes tool-independent principles of reading concordances.
2. **Develop a concordance software framework and new algorithms** that support a consistent concordance reading methodology: the **FlexiConc** library
3. **Conduct case studies** to illustrate the usefulness of the approach



Type '/' to search projects



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FlexiConc 0.1.17

```
pip install FlexiConc
```



[Latest version](#)

Released: Aug 4, 2025

A Python package to assist concordance reading

Navigation

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⬇️ [Download files](#)

Verified details ✓

Project description

Introduction to FlexiConc

FlexiConc is a Python package developed to facilitate and enhance the computational analysis of concordances, primarily intended for use in corpus linguistic research. The package is rooted in theoretical insights from the “Reading Concordances in the 21st Century” research project supported by the Arts and Humanities Research Council (AHRC) (grant references: AH/X002047/1 & AH/X002047/2) and the Deutsche Forschungsgemeinschaft (DFG) (grant reference: 508235423).

Hands-on: FlexiConc in CLiC basics

<https://clic-fiction.com/>

- We go back to the eyes in CLiC examples to think about extending the ways of looking at the data.
- With the help of various algorithms provided by FlexiConc.
- https://github.com/reading-concordances/teaching/blob/main/course/handouts/flexiconc_clic_handout_2026.pdf